

Exercise Corrections — Expected Results

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This document gives the expected R outputs, key results to look for, and interpretation criteria for each exercise. Exact numbers will depend on the dataset version used.

Exercise A — Functional space (Group: Neotropical + Nearctic)

Step 1 — Trait completeness:

Expected: most traits >85% complete. Eye diameter often most missing (~70%). After filtering, ~450–500 species remain. Two strongly correlated pairs likely: body length × body depth ($r \approx 0.7$); eye diameter × head length ($r \approx 0.65$). These represent the body size axis and head morphology axis.

Step 2 — PCA:

Expected: PC1 $\approx$ 40–50% variance, PC2 $\approx$ 15–25%. PC1 loads on body length, dorsal fin area, caudal peduncle — represents the **body size / pelagic vs benthic** axis. PC2 loads on eye size, mouth position — represents **visual predator vs substrate feeder**.

funspace plot: Neotropical region should be much larger and more spread. Overlap $\approx$ 0.05–0.15 (very low). This is consistent with Toussaint et al. (2016): Neotropical realm hosts disproportionate functional diversity.

Step 3 — Uniqueness:

Most unique species should be morphologically extreme: very elongated (electric eels, knifefish), very flattened (rays, armoured catfish), or very small with large eyes. Wilcoxon test: expect Neotropical species to be significantly more unique ($p < 0.05$). Biological reason: Neotropical contains many phylogenetically isolated lineages.

Exercise B — FD indices (Group: Palearctic + Ethiopian)

Step 1 — CWM:

Expected: CWM body length negatively correlated with temperature ($r \approx -0.3$ to -0.5). Linear model slope should be negative (larger fish in cooler basins = Bergmann's rule). Ethiopian basins: warm, smaller-bodied fish. Palearctic montane basins: cool, larger mean body length.

Step 2 — FD indices:

FRic–S correlation: expect $r \approx 0.6$ – 0.8 (strong). This is the confound — richer basins trivially show larger FRic. Between-realm comparison: Ethiopian FRic may be higher (more species) but Palearctic FDis may differ (different trait spread). FEve: expect lower in Palearctic (more constrained trait distribution). The key teaching point: always use SES when comparing FRic across richness gradients.

Step 3 — Trait–environment:

FDis vs temperature: may show positive correlation (warmer = more diverse strategies). If not significant, discuss why (confounded by richness, few data points).

Exercise C — Null models (Group: Oriental + Australian)

Step 1 — Observed FRic:

Strong FRic–S correlation expected ($r \approx 0.7$). This motivates the null model approach.

Step 2 — Null model and SES:

Expected SES distribution: most basins between -1.96 and $+1.96$ (random). Some outliers possible. Australian basins may show mild clustering ($SES \approx -1$ to -2) due to the depauperate, historically isolated fauna. Oriental basins may show near-random or slight overdispersion. Temperature vs SES: weak to moderate negative correlation (hotter = more clustering = more environmental filtering) possible.

Step 3 — Pattern distribution:

Expect most basins to be classified as 'random'. If clustering dominates: propose environmental filtering (harsh environments, few available niches). If overdispersion: propose competitive exclusion or habitat heterogeneity. Key teaching: the null model removes the richness effect — whatever remains is a signal of non-random assembly.

Group presentation evaluation criteria

Criterion	Poor (0-1)	Good (2-3)	Excellent (4-5)
Content accuracy	Major errors	Minor errors	Correct throughout
Figure quality	Unclear/missing	Readable	Clear, well-labelled
Ecological interpretation	None or wrong	Partial	Precise + references
Limitation	Not mentioned	Mentioned vaguely	Specific + discussed
Clarity (8 min)	Over/under time	Slightly off	Perfect timing

Total per group: 25 pts (5 criteria $\times$ 5 pts). Peer evaluation (optional): each student rates 1 other group on content + clarity.

Note: exact R outputs will depend on the dataset version provided. Check your own outputs before the session and note the key values (overlap, SES range, correlation coefficients) to guide the debrief.